Partitioning a disk

Partition table format

```
james@laptop:~$ sudo gdisk /dev/nvme1n1
GPT fdisk (gdisk) version 1.0.10

Partition table scan:
  MBR: protective
  BSD: not present
  APM: not present
  GPT: present

Found valid GPT with protective MBR; using GPT.
```

fdisk

fixed disk

for MBR partition table format (Master Boot Record)

`fdisk [disk]` **disk** that you want to partition

e.g. `fdisk /dev/hda`

`n`	create a partition
`d`	delete a partition
`l`	list partitions

example *sudo fdisk -l*

```
Disk /dev/nvme1n1: 931.51 GiB, 1000204886016 bytes, 1953525168 sectors
Disk model: CT1000P1SSD8
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 7BB2B1BC-47AE-43C4-854C-7CBE09E71029

Device              Start        End   Sectors   Size Type
/dev/nvme1n1p1         34      32767     32734    16M Microsoft reserved
/dev/nvme1n1p2      32768    1083391   1050624   513M EFI System
/dev/nvme1n1p3    1083392  977303551 976220160 465.5G Microsoft basic data
/dev/nvme1n1p4  977303552 1953521663 976218112 465.5G Linux filesystem


Disk /dev/nvme0n1: 931.51 GiB, 1000204886016 bytes, 1953525168 sectors
Disk model: Samsung SSD 970 EVO Plus 1TB
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: C7995594-A534-11EF-86C0-3448ED1BF768

Device            Start       End   Sectors   Size Type
/dev/nvme0n1p1     2048   2029567   2027520   990M Windows recovery environment
/dev/nvme0n1p2  2029576   2398223    368648   180M EFI System
/dev/nvme0n1p3  2398224   2660375    262152   128M Microsoft reserved
/dev/nvme0n1p4  2660376 978093383 975433008 465.1G Microsoft basic data
```

gdisk
for GPT table format
GUID **P**artition **T**able (Globally Unique Identifier)

n	create a partition
d	delete a partition
l	list partitions

parted
not in exam
for all table formats
- CLI or GUI
- resize partitions easily (provided their not being used)

Formatting a partition

Often called "making a filesystem"
- Involves writing low-level data structures to disk
- Linux can use these data structures to store and read files

low-level formatting
creates a structure on the disk
only low-level formatted when they are made

high-level formatting
creates a filesystem

Common Filesystem Types

ext2	**Second Extended File System** traditional Linux-native filesystem
ext4	**Fourth Extended File System** works with large disks improved performance
reiserFS	**journaling filesystem** traditional filesystems can crash while creating/modifying files and creates inconsistencies journaling keeps track of operations and has rollback options
FAT	**File Allocation Table** old filesystem supported by Microsoft DOS good for exchanging data between drives
NTFS	**New Technology File System** Windows filesystem Linux can read/modify but not write new files
ISO-9660	Filesystem for CD-ROMS

Creating a Filesystem

mkfs

example `mkfs -t ext3 /dev/sda6`
creates an ext3 filesystem on /dev/sda6

to learn about filesystem options
`man mkfs.[filesystem]`
e.g. *man mkfs.ext2*# Swap space

Swap space

an extension of memory
Linux can use a *swap file* which is a file that works the same way

```
james@laptop:~$ cat /proc/swaps
Filename                                Type            Size            Used            Priority
/swap.img                               file            16777212        0               -2
```